$$e_c(\mathbf{r}) = \|\widehat{C}_c(\mathbf{r}) - C(\mathbf{r})\|_2^2,$$

$$e_f(\mathbf{r}) = \|\widehat{C}_f(\mathbf{r}) - C(\mathbf{r})\|_2^2,$$

$$L = \sum_{\mathbf{r}\in\mathcal{R}} [e_c(\mathbf{r}) + e_f(\mathbf{r})]$$